# Shadows, Fibres and Secants of the Posterior Tower

Conditional expectations, Beck defects, and positive tropicalisation

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**Abstract**

The three-level posterior tower [1] has two candidate mixed distributive laws, and both fail some Beck axioms. This paper studies the failures through two operations on the tower's levels: *pooling*, which replaces a law by conditional expectations of it, and *tropicalisation*, which replaces probabilities by their large-deviation rates. It unifies and corrects three earlier manuscripts, on the Giry-monad formulation, on conditional shadows, and on fibres and secants. The changes are listed in appendix A.

*Algebra.* For finite partitions $A, B$ of a full-support probability space, five conditions are equivalent:

- $E_A E_B = E_B E_A$;
- $E_A E_B = E_{A \wedge B}$;
- $A$ and $B$ are conditionally independent given the meet $A \wedge B$;
- every connected component of the bipartite cell graph has a rank-one contingency block;
- every singular value of the normalised contingency matrix is 0 or 1.

The identity $E_A E_B = E_{A \vee B}$ holds only when $A = B$. In general $(E_A E_B)^n \to E_{A \wedge B}$ at the exact rate $c^{2n-1}$, so interchange holds laxly, through the meet.

*Shadows.* A coherent context determines a star-shaped probability space centred at the level-one predictive. Shadows are conditional expectations on this fixed space. They form a commutative band and satisfy interchange strictly. Shadows defined by label maps on laws do not compose as conditional expectations.

*Topology and order.* The barycentre has two continuous splittings, $e_\perp = \eta\mu$ and $e_\top = \mathsf{D}(\eta)\mu$. These form a left-zero band and bound every barycentre fibre in convex order. A coherent context is a three-step simplex-valued martingale. The shift law's level-one unit defect is the full convex-order diameter of a fibre; its multiplication defect is one conditional-expectation step.

*Categorical structure.* We construct the lifting functor from contexts to measure spaces in two versions and prove naturality of the shadows along measure-preserving maps. The double category is the quintet construction of a locally preordered 2-category, in which interchange is automatic. In this language each Beck defect computed so far is a single conditional-expectation step.

*Geometry.* Over a real closed non-Archimedean field the valuation is a monad morphism to the min-plus distribution monad. Classical barycentre fibres map onto tropical fibres. Positive mixtures map exactly onto tropical secants. The valuation factors through the Berkovich analytification, and on positive points the square commutes. The Bayes posterior tropicalises to the hard assignment, so the Voronoi walls of nearest-atom pooling are tropical walls. We also correct the Gaussian fusion square, the finite stochastic-to-convex functor and the moment-variety programme. Two scripts check every finite claim, with 30 and 8 checks, all passing.

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# 1 Introduction

## 1.1 The problem

A three-level posterior tower records three things: a predictive $\rho \in \mathsf{D}X$, a law $\Pi \in \mathsf{D}^2X$ over predictives, and a law $\Omega \in \mathsf{D}^3X$ over such laws. It also carries a shadow $\Sigma$ of labelled hypotheses. A candidate mixed distributive law on the tower may fail some Beck axioms, and the failures are comparisons between constructions, not merely numerical discrepancies. The motivating question is whether two ways of pooling information organise these comparisons, and whether the comparisons have a geometric meaning. Three logically separate tasks are involved:

(1) define the pooling operators (shadows) and prove their operator identities;

(2) build a category or double category in which the Beck comparisons are well-typed cells;

(3) locate the comparisons geometrically: in barycentre fibres, in convex order, and after tropicalisation, in secant geometry.

None of these implies another, and the paper keeps them apart.

## 1.2 Guiding principles

(i) *Compose operators on a fixed probability space.* Composing measure-valued maps that recompute their partitions after each step is a different operation. Most earlier difficulties came from mixing the two.

(ii) *Conditional expectations interact through the meet, never the join.*

(iii) *Probability tropicalises cleanly because it is positive.* Leading terms of positive series cannot cancel, so the valuation is a semiring homomorphism. Every geometric statement in the paper rests on this.

### 1.3 Main results

(A) **Commutation** (theorem 3.3): five equivalent conditions, a spectral and a combinatorial criterion among them. The join identity holds only for equal $\sigma$-algebras (proposition 3.4).

(B) **Rate and lax interchange** (theorem 3.6 and corollary 3.8).

(C) **Shadow algebra** (theorem 4.9): the canonical shadows of a context form a commutative band.

(D) **Fibres and martingale** (propositions 5.1 and 5.2 and theorem 5.3).

(E) **Beck defects located** (propositions 5.4 and 7.2).

(F) **Lifting functor and naturality** (propositions 4.6, 6.5 and 6.6); quintet double category (proposition 6.10).

(G) **Tropical tower** (theorem 11.3), **fibre lifting** (theorem 11.6), **tropical secants** (theorem 11.9) and the **Berkovich square** (theorem 11.11).

(H) **Zero-temperature shadows and walls** (propositions 12.1 and 12.2).

### 1.4 Status

| Statement | Status |
|---|---|
| operator commutation, rate, lax interchange | proved, finite and general |
| canonical shadows, band, naturality | proved |
| fibres, martingale, location of the (U) and (M) defects | proved |
| lifting functor, quintet double category | proved |
| a 2-category in which $\mathsf{D}$, $\mathsf{C}_3$ and the laws act monotonically | open (open problem 6.11) |
| tropical tower, fibre lifting, tropical secants, Berkovich square | proved |
| tropical identifiability inside a model | open (open problem 12.5) |
| wall-crossing formula for Gaussian mixture towers | proposed (conjecture 12.4) |
| Beck defects as Čech classes | no construction; open (open problem 12.7) |

### 1.5 Conventions

$X$ and $B$ denote finite sets, and $\mathsf{D}$ is the finite distribution monad with $\eta_X(x) = \delta_x$ and $\mu_X(\Pi) = \sum_\rho \Pi(\rho)\rho$. We identify $\mathsf{D}B$ with the simplex $\Delta_B \subset \mathbb{R}^B$, with the $\ell^1$ metric. A *finite probability space* $(I, p)$ has $p(i) > 0$ for all $i$, and its $\sigma$-algebras are partitions. For a partition $A$, $E_A$ is conditional expectation, acting coordinatewise on $\mathbb{R}^X$-valued functions: $(E_A Z)(i) = \sum_{j \in A(i)} p(j)Z(j)/p(A(i))$, where $A(i)$ is the cell of $i$. In the tower paper [1], the letters $L$ and $R$ also denote the two sides of the (Cm) diagram. Here $L_c$ and $R_c$ always denote shadows (definition 4.8).

# 2 The posterior tower

## 2.1 Levels and contexts

For a finite set $B$, the iterated distribution spaces $\mathsf{D}B, \mathsf{D}^2B, \mathsf{D}^3B, \ldots$ carry barycentre maps satisfying the monad laws. A hypothesis is $h = (\theta, r) \in \mathsf{Hyp}B = \Theta \times \mathsf{D}B$, with predictive $q(h) = r$. $\mathsf{Hyp}$ is a functor and $q$ is natural [1, Def. 3.1]. The shadow space is $\mathsf{Sh}(B) = \mathsf{D}(\mathsf{Hyp}B)$. A context is $c = (\rho, \Pi, \Omega, \Sigma)$ with $\rho \in \mathsf{D}B$, $\Pi \in \mathsf{D}^2B$, $\Omega \in \mathsf{D}^3B$ and $\Sigma \in \mathsf{Sh}B$.

**Definition 2.1** (Coherent context)**.** $c$ is *coherent* if

$$\mu(\Pi) = \rho, \qquad \mu(\Omega) = \Pi, \qquad \mathsf{D}(q)(\Sigma) = \Pi. \tag{1}$$

**Remark 2.2** (Typing)**.** The third condition is meaningful because $q$ lands in $\mathsf{D}B$, the underlying space of $\Pi$. Any variant of the shadow component must come with its own projection to $\mathsf{D}B$ and its action on maps $f\colon B \to B'$. For $\mathsf{Hyp} = \Theta \times \mathsf{D}(-)$ this action is $\mathsf{Hyp}(f)(\theta, r) = (\theta, f_*r)$.

## 2.2 The absolute tower and the two laws

The absolute tower $\mathsf{C}_3X = X \times \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{Sh}X$ is a comonad, with comultiplication $\delta(x, c) = ((x, c), \mathsf{E}_3(j_c)(c))$, where $j_c(y) = (y, c)$ [1, §6]. Applying it at $\mathsf{D}X$ gives $\mathsf{C}_3(\mathsf{D}X) = \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{D}^4X \times \mathsf{Sh}(\mathsf{D}X)$. For a sample $M = \sum_i a_i\delta_{(x_i,c_i)}$:

- the *shift law* $\lambda^\sigma$ keeps each sample's context entries, moving them one level up;
- the *averaging law* $\bar{\lambda}$ pushes contexts through the unit and averages them.

## 2.3 Beck axioms

A mixed distributive law $\lambda\colon \mathsf{DC} \Rightarrow \mathsf{CD}$ must satisfy the unit (U), counit (Co), multiplication (M) and comultiplication (Cm) diagrams [1, Def. 2.1]. The shift law satisfies (Co) and fails (U), (M) and (Cm). The averaging law satisfies (U), (Co) and (M) and fails (Cm) [1, Thms. 8.3, 8.6, 8.7]. The base-relative tower has a strict law [1, Thm. 9.8]. A failure produces two parallel composites with the same source and target, and the comparison between them is the *Beck defect*.

# 3 Algebra of conditional expectations

## 3.1 The partition lattice

Partitions of $I$ form a lattice under refinement. The *join* $A \vee B$ has cells $a \cap b$. The *meet* $A \wedge B$ is the finest common coarsening. As $\sigma$-algebras, $A \vee B = \sigma(A \cup B)$ and $A \wedge B = A \cap B$.

**Lemma 3.1** (The meet is $\pi_0$ of a bipartite graph)**.** *Let $G(A, B)$ be the bipartite graph on the cells of $A$ and of $B$, with an edge $a$–$b$ whenever $a \cap b \neq \emptyset$. The cells of $A \wedge B$ are the unions over the connected components of $G(A, B)$, and $L^2(A) \cap L^2(B) = L^2(A \wedge B)$.*

*Proof.* A set lies in $A \cap B$ iff it is a union of $A$-cells and a union of $B$-cells, i.e. iff it is closed under the edge relation. A function lies in $L^2(A) \cap L^2(B)$ iff it is constant on cells of both partitions, i.e. iff it is constant on components. □

### 3.2 The commutation theorem

Write $N_{ab} = p(a \cap b)$ for the contingency matrix and $K = D_A^{-1/2} N D_B^{-1/2}$, with $D_A = \mathrm{diag}(p(a))$ and $D_B = \mathrm{diag}(p(b))$.

**Lemma 3.2** (A sufficient pair)**.** *On any probability space, let $C \subseteq A \cap B$ be sub-$\sigma$-algebras such that $A$ and $B$ are conditionally independent given $C$. Then $E_A E_B = E_C = E_B E_A$, and $C = A \cap B$ up to null sets.*

*Proof.* For $V \in L^2(B)$ and $U \in L^2(A)$, conditional independence gives $\mathbb{E}[UV] = \mathbb{E}[\mathbb{E}[U \mid C]\mathbb{E}[V \mid C]] = \mathbb{E}[U\,\mathbb{E}[V \mid C]]$, using $C \subseteq A$. So $E_A V = E_C V$, and hence $E_A E_B = E_C E_B = E_C$, using $C \subseteq B$. The other order is symmetric. The range of $E_C$ is $L^2(C)$, and that of the commuting product is $L^2(A) \cap L^2(B) = L^2(A \cap B)$. □

**Theorem 3.3** (Commutation)**.** *For partitions $A, B$ of a finite probability space the following are equivalent:*

*(i)* $E_A E_B = E_B E_A$*;*

*(ii)* $E_A E_B = E_{A \wedge B}$*;*

*(iii)* *$A$ and $B$ are conditionally independent given $A \wedge B$;*

*(iv)* *on each component $\gamma$ of $G(A, B)$, the block $N|_\gamma$ has rank one, i.e. $p(a \cap b)p(\gamma) = p(a)p(b)$ for all cells $a, b \subseteq \gamma$;*

*(v)* *every singular value of $K$ is $0$ or $1$.*

*The equivalences (i)⇔(ii)⇔(iii) hold on every probability space, with $A \wedge B$ read as $A \cap B$. The value $1$ in (v) has multiplicity equal to the number of components. Nestedness ($A \subseteq B$ or $B \subseteq A$) implies (iv).*

*Proof.* (iii)⇒(ii) is lemma 3.2. (ii)⇒(i): $E_{A \wedge B}$ is self-adjoint, so $E_B E_A = (E_A E_B)^* = E_A E_B$. (i)⇒(ii): the product of commuting orthogonal projections is the projection onto the intersection of their ranges, which is $L^2(A \wedge B)$ by lemma 3.1. (ii)⇒(iii): with $C = A \wedge B$, $\mathbb{E}[UV \mid C] = \mathbb{E}[U\,E_A V \mid C] = \mathbb{E}[U\,E_C V \mid C] = E_C U \cdot E_C V$, since $E_A V = E_A E_B V = E_C V$ is $C$-measurable.

(iii)⇔(iv): on a finite space it suffices to test indicators. Cells lie inside single components, so conditional independence given $C$ reads $p(a \cap b \mid \gamma) = p(a \mid \gamma)p(b \mid \gamma)$. This says $N|_\gamma$ is the outer product of its margins divided by its total, which for a nonnegative matrix is the same as rank one.

(ii)⇔(v): in the orthonormal bases $1_a/\sqrt{p(a)}$ and $1_b/\sqrt{p(b)}$, $E_A|_{L^2(B)}$ has matrix $K$, whose singular values are the cosines of the principal angles between $L^2(A)$ and $L^2(B)$. The value 1 occurs exactly on $L^2(A \wedge B)$. Condition (ii) holds iff all other angles are $\pi/2$. If $A \subseteq B$, every $A$-cell is a component and (iv) is trivial. □

Check A1 confirms the five-way equivalence on 600 random instances, 449 commuting and 151 not.

**Proposition 3.4** (The join identity)**.** *$E_A E_B = E_{A \vee B}$ holds iff $A = B$ up to null sets.*

*Proof.* The range of $E_A E_B$ lies in $L^2(A)$. If it equals $L^2(A \vee B) \supseteq L^2(B)$, then $B \subseteq A$, so $E_A E_B = E_B$ and $E_{A \vee B} = E_A$, whence $A = B$. □

**Example 3.5** (A non-commuting pair)**.** Let $I = \{1,2,3,4\}$ with $p = (\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8})$, $Z(i) = i$, $A_L = \{\{1,2\},\{3,4\}\}$ and $A_R = \{\{1,3\},\{2,4\}\}$. Then

$$(E_{A_R}E_{A_L}Z)(1) = \tfrac{3}{4}\cdot\tfrac{5}{4} + \tfrac{1}{4}\cdot\tfrac{15}{4} = \tfrac{15}{8}, \qquad (E_{A_L}E_{A_R}Z)(1) = \tfrac{3}{4}\cdot\tfrac{3}{2} + \tfrac{1}{4}\cdot\tfrac{7}{2} = 2.$$

$G(A_L, A_R)$ is a 4-cycle, so the meet is trivial. The block $N = \left(\begin{smallmatrix} 3/8 & 1/8 \\ 1/8 & 3/8 \end{smallmatrix}\right)$ has rank two, and $K$ has singular values 1 and $\frac{1}{2}$ (check A3).

### 3.3 Worked regimes

- *Committed* ($A = B$): the operators coincide. This is a consistency check only.
- *Inherited* ($A \subseteq B$ or $B \subseteq A$): the operators commute with product the coarser one.
- *General*: commutation holds iff (iv) of theorem 3.3 holds. Adding labels does not by itself produce conditional independence. Example 3.5 shows the generic outcome.

### 3.4 The general case: rate and lax interchange

**Theorem 3.6** (Alternating projections, exact rate)**.** *Let $c$ be the cosine of the Friedrichs angle between $L^2(A)$ and $L^2(B)$. In the finite case, $c$ is the largest singular value of $K$ strictly below 1, or 0 if there is none. Then*

$$\|(E_AE_B)^n - E_{A\wedge B}\| = c^{2n-1} \qquad (n \geq 1).$$

*On a general probability space the identity holds with the Friedrichs cosine. Then $c < 1$ iff $L^2(A) + L^2(B)$ is closed. Otherwise $(E_AE_B)^n \to E_{A\wedge B}$ only strongly, and the convergence can be arbitrarily slow.*

*Proof.* The identity is the Kayalar–Weinert theorem [14], with $L^2(A) \cap L^2(B) = L^2(A \wedge B)$. Strong convergence is von Neumann's theorem [19], and arbitrarily slow convergence is in [5]. □

Check A4 verifies the rate to $10^{-9}$ for $n \leq 6$ on 157 non-commuting pairs.

**Definition 3.7** (Convex order)**.** For finitely supported laws $Q, Q'$ on a convex set, $Q \leq_{\text{cx}} Q'$ if $\int \varphi\, dQ \leq \int \varphi\, dQ'$ for all convex $\varphi$. By Strassen's theorem [18] this holds iff $Q = \text{Law}(\mathbb{E}[U' \mid G])$ for some $U' \sim Q'$ and some sub-$\sigma$-algebra $G$.

**Corollary 3.8** (Lax interchange)**.** $E_{A\wedge B}E_AE_B = E_{A\wedge B} = E_{A\wedge B}E_BE_A$. *Hence*

$$\text{Law}(E_{A\wedge B}Z) \leq_{\text{cx}} \text{Law}(E_AE_BZ) \leq_{\text{cx}} \text{Law}(E_BZ) \leq_{\text{cx}} \text{Law}(Z),$$

*and symmetrically through $E_BE_AZ$. The laws $\text{Law}((E_AE_B)^nZ)$ decrease in convex order, and*

$$W_2\big(\text{Law}((E_AE_B)^nZ), \text{Law}(E_{A\wedge B}Z)\big) \leq c^{2n-1}\|Z - E_{A\wedge B}Z\|_{L^2(p)}.$$

*Proof.* The tower property gives the first identity. Each displayed law is that of a conditional expectation of the next on the same space. For the rate, $(E_AE_B)^n - E_{A\wedge B} = ((E_AE_B)^n - E_{A\wedge B})(I - E_{A\wedge B})$; apply theorem 3.6 and use the identity coupling to bound $W_2$. □

So without nestedness or conditional independence, interchange does not hold strictly. It holds through the cospan $E_AE_B \Rightarrow E_{A\wedge B} \Leftarrow E_BE_A$ of convex-order comparisons. These are identities exactly under theorem 3.3, and they shrink at the rate of theorem 3.6.

### 3.5 Conditioning on a third variable

**Proposition 3.9.** *Let $X_L \perp X_R \mid H$ on a finite space with full support. Then:*

*(a)* $E_{\sigma(X_L,H)}E_{\sigma(X_R,H)} = E_{\sigma(H)} = E_{\sigma(X_R,H)}E_{\sigma(X_L,H)}$;

*(b)* $E_{\sigma(X_L)}$ *and* $E_{\sigma(X_R)}$ *commute iff* $X_L \perp X_R$ *marginally. With full support the meet is trivial, so this fails as soon as both channels depend on* $H$.

*Proof.* (a) is lemma 3.2 with $C = \sigma(H)$. (b) With full support, $G(\sigma(X_L), \sigma(X_R))$ is complete bipartite, hence connected; apply theorem 3.3. □

Conditional independence given a third variable is therefore not a commutation hypothesis. The conditioning variable must be built into both $\sigma$-algebras (check A7).

## 4 Shadows

### 4.1 Label-defined shadows on laws

Let $S = \mathsf{D}B$ with its Borel $\sigma$-algebra, and let $a\colon S \to I_a$ be a measurable *label map* with finite codomain. For $\nu \in \mathsf{D}(S)$ with identity variable $Z$, put $m_a = \mathbb{E}_\nu[Z \mid \sigma(a)]$ and define the label shadow $P_a(\nu) = (m_a)_*\nu$.

**Proposition 4.1** (Properties of a single label shadow)**.** *(a)* $\int \rho\, dP_a(\nu) = \int \rho\, d\nu$, *and* $P_a(\nu) \leq_{\mathrm{cx}} \nu$.

*(b) If* $a(m_a(\rho)) = a(\rho)$ *for* $\nu$*-a.e.* $\rho$ *(*label preservation*), then* $P_a(P_a\nu) = P_a\nu$. *Without label preservation, the first barycentre can move a point into another label class, and idempotence can fail.*

*Proof.* (a) is the tower property together with Strassen's theorem. (b) Write $Y = m_a(Z)$. Then $Y$ is $\sigma(a(Z))$-measurable, and label preservation gives $a(Y) = a(Z)$, so $\mathbb{E}[Y \mid a(Y)] = Y$. □

**Proposition 4.2** (Composites of label shadows are single poolings)**.** *For label maps* $a, b$,

$$P_b(P_a(\nu)) = \mathrm{Law}\left(E_{A'}Z\right), \qquad A' = \sigma(b \circ m_a) \subseteq \sigma(a).$$

*So* $P_bP_a$ *pools once, onto a coarsening of the* first *partition. The equation* $P_bP_a = P_aP_b$ *compares coarsenings of two different partitions. It is not the operator identity* $E_AE_B = E_BE_A$, *and theorem 3.3 does not govern it.*

*Proof.* $m_a$ is constant on $\sigma(a)$-cells, so $b \circ m_a$ is too. Pooling the atoms of $P_a\nu$ along $b$ merges whole cells, and a barycentre of barycentres is the barycentre of the union. □

**Example 4.3** (Threshold labels)**.** On $\mathsf{D}\{0,1\} \cong [0,1]$, let $\nu$ be uniform on $\{0.1, 0.3, 0.7, 0.9\}$, $a = [s \geq \frac{1}{2}]$ and $b = [s \geq \frac{1}{4}]$. Then $P_b(P_a\nu) = \frac{1}{2}\delta_{0.2} + \frac{1}{2}\delta_{0.8}$, whereas $P_a(P_b\nu) = \frac{1}{4}\delta_{0.1} + \frac{3}{4}\delta_{19/30}$. The fixed-space operators also fail to commute, but with different values (checks C2, C3).

### 4.2 Labelled random measures

**Definition 4.4.** A *labelled random measure* over $X$ is a triple $(I, p, Z)$: a finite probability space and a map $Z\colon I \to \mathsf{D}X$. A morphism $f\colon (I,p,Z) \to (I',p',Z')$ is a map with $f_*p = p'$ and $Z' \circ f = Z$. The functor $\mathrm{Law}(I,p,Z) = Z_*p \in \mathsf{D}^2X$ forgets the labels.

**Proposition 4.5** (Shadows on a fixed space)**.** *For partitions $A, B$ of $I$:*

*(a) $E_A$ is idempotent, preserves $\mathbb{E}Z$, and satisfies $\mathrm{Law}(E_AZ) \le_{\mathrm{cx}} \mathrm{Law}(Z)$;*

*(b) composition of shadows is composition of conditional expectations;*

*(c) $\mathrm{Law}(E_AE_BZ) = \mathrm{Law}(E_BE_AZ)$ for all $Z$ iff (i)–(v) of theorem 3.3 hold (when $|X| \ge 2$).*

*Proof.* (a) and (b) are immediate. For "only if" in (c): the energy $\int \|v\|^2\, d\,\mathrm{Law}(\cdot)$ is a function of the law. Write $Z = \rho_0 + Y$ with $\rho_0$ in the relative interior and $Y$ small with zero coordinate sums. Then $\|E_AE_BY\| = \|E_BE_AY\|$ on an open set, hence everywhere, and polarisation gives $S := E_BE_AE_B = E_AE_BE_A$. The range of $S$ lies in $L^2(M)$ with $M = A \wedge B$, and $S$ is self-adjoint, so $S = E_MSE_M = E_M$. For $Y \in L^2(B)$ this reads $\|E_AY\| = \|E_MY\|$, so $E_AY = E_MY$ by Pythagoras, which is (ii). □

**Proposition 4.6** (Naturality)**.** *Let $f\colon (I,p) \to (I',p')$ be measure preserving, and let $A = f^{-1}A'$, $B = f^{-1}B'$. Then $E_A(Z' \circ f) = (E_{A'}Z') \circ f$, and likewise for $B$.*

*Proof.* $f^{-1}(a')$ is a cell of $A$ of mass $p'(a')$, and the average of $Z' \circ f$ over it is the average of $Z'$ over $a'$ (check G2). □

### 4.3 The star space of a context and the canonical shadows

**Definition 4.7** (Star space)**.** For a coherent context $c$, let $\kappa(\theta \mid r) = \Sigma(\theta, r)/\Pi(r)$. The *star space* of $c$ has points $(\widetilde{\Pi}, r, \theta, x)$ with weights

$$p_c(\widetilde{\Pi}, r, \theta, x) = \Omega(\widetilde{\Pi})\, \widetilde{\Pi}(r)\, \kappa(\theta \mid r)\, r(x),$$

restricted to positive weights. Write $\tilde{r}$ for the coordinate $r$ and $Z = \delta_x$.

Given $\tilde{r}$, the three *leaves* $\widetilde{\Pi}$, $\theta$ and $x$ are independent. The marginals recover the context: $\mathrm{Law}(\widetilde{\Pi}) = \Omega$, $\mathrm{Law}(\tilde{r}) = \Pi$, $\mathrm{Law}(\theta, \tilde{r}) = \Sigma$ and $\mathrm{Law}(x) = \rho$ (check B1). The context itself does not couple $\Omega$ with $\Sigma$; the conditional product over $\tilde{r}$ is the canonical coupling.

**Definition 4.8** (Canonical shadows)**.** For a set $T \subseteq \{\mathrm{lev2}, \mathrm{hyp}, \mathrm{obs}\}$ of leaves, let

$$G_T = \sigma(\tilde{r},\ \text{leaves in } T), \qquad S_T = E_{G_T}.$$

The *left shadow* is $L_c = S_{\{\mathrm{lev2}\}}$ and the *right shadow* is $R_c = S_{\{\mathrm{hyp}\}}$.

**Theorem 4.9** (The shadow algebra is a commutative band)**.** *$S_TS_U = S_{T\cap U} = S_US_T$ for all leaf sets $T, U$. In particular $L_cR_c = R_cL_c = E_{\sigma(\tilde{r})}$.*

*Proof.* $G_{T\cap U} \subseteq G_T \cap G_U$. The leaves in $T \setminus U$ and in $U \setminus T$ are independent given $\tilde{r}$ and the leaves in $T \cap U$, so lemma 3.2 applies. □

Check B2 verifies all $8 \times 8$ products on 60 random coherent contexts. The homomorphism $T \mapsto S_T$ need not be injective: if $\theta$ is a function of $\tilde{r}$, then $R_c = S_\emptyset$.

**Proposition 4.10** (Dropping the centre)**.** *The label-only shadows $E_{\sigma(\widetilde{\Pi})}$ and $E_{\sigma(\theta)}$ commute iff theorem 3.3 holds for them. They fail to commute in 40 of 60 random contexts (check B3).*

**Proposition 4.11** (Markov absorption)**.** *For $Z = \delta_x$: (a) $R_c Z = E_{\sigma(\tilde{r})} Z = \tilde{r}$; (b) $E_{\sigma(\theta)} Z = \sum_r \Sigma(r \mid \theta)\, r$, and in general $E_{\sigma(\tilde{r})} E_{\sigma(\theta)} Z \neq E_{\sigma(\theta)} Z$ (59 of 60 random contexts, check B5).*

*Proof.* (a) holds because $x \perp \theta \mid \tilde{r}$; (b) is a direct computation. □

The hypothesis level is bookkeeping: given the predictive, it does not move the prediction.

# 5 Topology and order: how the tower acts

## 5.1 Two idempotents

The unit laws $\mu \circ \eta_{\mathsf{D}} = \mathrm{id} = \mu \circ \mathsf{D}(\eta)$ give one retraction $\mu \colon \mathsf{D}^2 X \to \mathsf{D}X$ with two sections, and hence two split idempotents on $\mathsf{D}^2 X$:

$$e_\perp = \eta_{\mathsf{D}X} \circ \mu \;\; (\Pi \mapsto \delta_{\mu\Pi}), \qquad e_\top = \mathsf{D}(\eta_X) \circ \mu \;\; (\Pi \mapsto \textstyle\sum_x (\mu\Pi)(x)\, \delta_{\delta_x}).$$

**Proposition 5.1** (Two idempotents)**.** *(a) $e_\perp$ and $e_\top$ are idempotent, natural in $X$, and split through the same retraction $\mu$.*

*(b) They form a left-zero band: $e_\perp e_\top = e_\perp$ and $e_\top e_\perp = e_\top$.*

*(c) $e_\top$ is affine; $e_\perp$ is not.*

*(d) For $W_1$ built from $\ell^1$: $\mu$ is 1-Lipschitz, $\eta$ and $\mathsf{D}(\eta)$ are isometries, and $e_\perp$, $e_\top$ are 1-Lipschitz, hence weakly continuous.*

*Proof.* (a) These are the unit laws and naturality of $\eta$ and $\mu$. (b) $\eta\mu\mathsf{D}(\eta)\mu = \eta\mu$ and $\mathsf{D}(\eta)\mu\eta\mu = \mathsf{D}(\eta)\mu$. (c) $\mu$ and $\mathsf{D}(\eta)$ are affine. $\eta$ is not, since $\delta_{s\rho+(1-s)\rho'} \neq s\delta_\rho + (1-s)\delta_{\rho'}$. (d) $r \mapsto \langle g, r\rangle$ is 1-Lipschitz for $g \in [-1,1]^X$, so $\|\mu\Pi - \mu\Pi'\|_1 \leq W_1(\Pi, \Pi')$. Also $W_1(\delta_\rho, \delta_{\rho'}) = \|\rho - \rho'\|_1$, and distinct vertices are at distance 2, so $W_1(\mathsf{D}(\eta)\rho, \mathsf{D}(\eta)\rho') = \|\rho - \rho'\|_1$. □

The pooling idempotent of the tower paper is $e = e_\perp$ [1, §13.6], [2].

## 5.2 Barycentre fibres

**Proposition 5.2** (Fibres are convex-order intervals)**.** *For $\rho \in \mathsf{D}X$, let $F_\rho = \mu^{-1}(\rho)$ be the set of Borel probability measures on $\Delta_X$ with barycentre $\rho$. Then:*

*(a) $F_\rho$ is convex and weakly compact, and its finitely supported elements are dense in it;*

*(b) $e_\perp \Pi = \delta_\rho \leq_{\mathrm{cx}} \Pi \leq_{\mathrm{cx}} \mathsf{D}(\eta)\rho = e_\top \Pi$ for every $\Pi \in F_\rho$;*

*(c) $\mathsf{D}(\eta)\rho$ is the unique element of $F_\rho$ supported on the vertices.*

*Proof.* (a) Prokhorov's theorem and continuity of $\mu$ give compactness; $F_\rho$ is convex because $\mu$ is affine. For density, partition $\Delta_X$ into cells of diameter $< \varepsilon$ and pool $\Pi$ within each cell. This keeps the barycentre and moves $\Pi$ by less than $\varepsilon$ in $W_1$. (b) $\rho \mapsto r \sim \Pi$ and $r \mapsto \delta_x$ with probability $r(x)$ are martingale couplings (check B6). (c) Vertices are affinely independent. This is the Choquet–Meyer uniqueness of maximal measures on a simplex [17]. □

### 5.3 A coherent context is a martingale

**Theorem 5.3** (The tower as a martingale)**.** *On the star space let $\mathcal{F}_0$ be trivial, $\mathcal{F}_1 = \sigma(\widetilde{\Pi})$, $\mathcal{F}_2 = \sigma(\widetilde{\Pi}, \tilde{r})$ and $\mathcal{F}_3 = \sigma(\widetilde{\Pi}, \tilde{r}, x)$, and put $M_k = \mathbb{E}[Z \mid \mathcal{F}_k]$. Then $(M_k)$ is a $\Delta_X$-valued martingale with*

$$\mathrm{Law}(M_0) = \delta_\rho, \quad \mathrm{Law}(M_1) = \mathsf{D}(\mu)\Omega, \quad \mathrm{Law}(M_2) = \Pi, \quad \mathrm{Law}(M_3) = \mathsf{D}(\eta)\rho,$$

*a chain $\delta_\rho \leq_{\mathrm{cx}} \mathsf{D}(\mu)\Omega \leq_{\mathrm{cx}} \Pi \leq_{\mathrm{cx}} \mathsf{D}(\eta)\rho$ in $F_\rho$. The hypothesis branch does not move it: $\mathbb{E}[Z \mid \theta, \tilde{r}] = M_2$.*

*Proof.* $M_0 = \rho$, $M_1 = \mu(\widetilde{\Pi})$, $M_2 = \tilde{r}$ and $M_3 = \delta_x$. The last statement is proposition 4.11 (check B4). □

So the levels of the tower are the successive conditional expectations of the observation along one filtration. Coherence is the martingale property in law. Nested shadows land on levels and commute.

### 5.4 The Beck defects located

**Proposition 5.4** (Where the defects of $\lambda^\sigma$ sit)**.** *At level one:*

*(a) the two sides of (U) at $(x, c)$ are $e_\perp \Pi = \delta_\rho$ and $e_\top \Pi = \mathsf{D}(\eta)\rho$, so the unit defect is the full convex-order diameter of $F_\rho$;*

*(b) for $M' = \sum_\ell \nu_\ell \delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k} \delta_{(x_{\ell k}, c_{\ell k})}$, on the sample-index space $I = \{(\ell, k)\}$ with $p = \nu_\ell a_{\ell k}$ and $W(\ell, k) = \rho_{\ell k}$, the two sides of (M) are $\mathrm{Law}(W)$ ($\lambda$-first) and $\mathrm{Law}(\mathbb{E}[W \mid \ell])$.*

*Proof.* This is [1, Prop. 13.8] read in the present notation, and it is verified with the tower code's own $\lambda^\sigma$, $\mathsf{C}_3$, $\eta$ and $\mu$ (check B7). □

The shift law pools after the unit and spreads after the multiplication. Every order that reads conditional expectation monotonically therefore orients the two defects oppositely [1, Prop. 13.9, Thm. 13.15, Prop. 13.17].

### 5.5 Continuity

**Proposition 5.5.** *(a) On a fixed index with fixed partitions, $(p, Z) \mapsto \mathrm{Law}(E_A Z)$ is continuous for full-support $p$, and $W_1(\mathrm{Law}(E_A Z), \mathrm{Law}(E_A Z')) \leq \|Z - Z'\|_{L^1(p)}$. (b) Nearest-atom shadows are Borel but not continuous. With atoms $(\frac{2}{3}, \frac{1}{3})$ and $(\frac{1}{3}, \frac{2}{3})$, two inputs at $W_1$-distance $\frac{1}{1000}$ have outputs at distance $\frac{501}{2000}$ (check C4).*

*Proof.* (a) Cell averages are continuous where the cell masses are positive, and conditional expectation is an $L^1$ contraction. (b) A point crossing the Voronoi wall changes cells. □

## 6 Categorical structure

### 6.1 Why a separate construction is needed

The operator results act on measures and random variables; context maps act on contexts. A double category needs objects, horizontal and vertical arrows with associative compositions, squares with typed boundaries, identity squares, two compositions of squares, and the interchange law for composable grids. Expressions such as $g \circ L$ or $R \circ f$ are undefined unless a functor connects the two layers. This section builds that functor and then the double category.

## 6.2 The context category

**Definition 6.1** (Context category)**. Ctx** has as objects the coherent contexts $c$ on finite bases $B_c$. A morphism $f\colon c \to c'$ is a map $f\colon B_c \to B_{c'}$ with

$$c' = \big(f_*\rho,\ \mathsf{D}^2 f(\Pi),\ \mathsf{D}^3 f(\Omega),\ \mathsf{Sh} f(\Sigma)\big).$$

Composition and identities are those of maps of finite sets.

**Proposition 6.2. Ctx** *is a category, and pushing forward along $f$ preserves coherence.*

*Proof.* $\mathsf{D}^k$ and $\mathsf{Sh}$ are functors. Coherence is preserved by naturality of $\mu$ ($\mu \circ \mathsf{D}^2 f = \mathsf{D} f \circ \mu$) and of $q$. □

## 6.3 The lifting functor, in two versions

**Remark 6.3** (Two readings of the level-two component)**.** A morphism $f\colon c \to c'$ can carry either of two kinds of level-two data:

(i) a specified map $f_\Pi\colon \mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c'})$, with $(g \circ f)_\Pi = g_\Pi \circ f_\Pi$ imposed as an axiom (*component-data version*);

(ii) a base map $f\colon B_c \to B_{c'}$, with induced action $\mathsf{D}^2(f)$ (*base-map version*, as in definition 6.1).

The two agree when $f_\Pi = \mathsf{D}^2(f)$. This is a coherence condition, not automatic.

**Definition 6.4** (Lifting functor)**.** Define $F(c) = \mathsf{D}^2(B_c)$, with $F(f) = f_\Pi$ in the component-data version and $F(f) = \mathsf{D}^2(f)$ in the base-map version.

**Proposition 6.5** (Functoriality, component-data version)**.** *If* $(\mathrm{id}_c)_\Pi = \mathrm{id}$ *and* $(g \circ f)_\Pi = g_\Pi \circ f_\Pi$*, then $F$ is a functor.*

*Proof.* $F(\mathrm{id}_c) = (\mathrm{id}_c)_\Pi = \mathrm{id}_{F(c)}$ and $F(g \circ f) = (g \circ f)_\Pi = g_\Pi \circ f_\Pi = F(g) \circ F(f)$, and all three maps have type $\mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c''})$. □

**Proposition 6.6** (Functoriality, base-map version)**.** *In the base-map version $F$ is a functor.*

*Proof.* $\mathsf{D}^2$ is a functor. □

**Remark 6.7.** In the component-data version $f_\Pi$ is already a map between level-two spaces. Applying $\mathsf{D}^2$ to it again would give a map $\mathsf{D}^4(B_c) \to \mathsf{D}^4(B_{c'})$ of the wrong type, so $F(f) = f_\Pi$ is the type-correct choice there.

## 6.4 Shadows along context maps

The canonical shadows live on star spaces, not on $F(c)$. The star construction is functorial on a large subcategory.

**Proposition 6.8** (Star spaces are functorial)**.** *Let $f\colon c \to c'$ be a morphism of* **Ctx** *such that* $\mathsf{D} f$ *is injective on* $\operatorname{supp}\Pi$*. Then*

$$\tilde{f}(\widetilde{\Pi}, r, \theta, x) = \big(\mathsf{D}^2 f(\widetilde{\Pi}),\ \mathsf{D} f(r),\ \theta,\ f(x)\big)$$

*maps the star space of $c$ measure-preservingly onto the star space of $c'$. It pulls back $G_T$ to $G_T$ for $T \subseteq \{\mathrm{lev2}, \mathrm{hyp}\}$. Hence $L_c$ and $R_c$ are natural: $L_c(Z' \circ \tilde{f}) = (L_{c'} Z') \circ \tilde{f}$, and likewise for $R$.*

*Proof.* Summing $p_c$ over the fibre of $\tilde{f}$ gives $\Omega(\widetilde{\Pi})\widetilde{\Pi}(r)\kappa(\theta \mid r)\sum_{x\in f^{-1}(x')} r(x)$. By injectivity on atoms, each factor equals its primed counterpart. Injectivity also gives $\tilde{f}^{-1}\sigma(\mathsf{D}f(\tilde{r})) = \sigma(\tilde{r})$ and $\tilde{f}^{-1}\sigma(\mathsf{D}^2 f(\widetilde{\Pi})) = \sigma(\widetilde{\Pi})$, while $\theta$ is unchanged. Naturality is proposition 4.6 (check G3, on 36 context maps). For $T \ni \mathrm{obs}$ only $\tilde{f}^{-1}G'_T \subseteq G_T$ holds, because $f$ may merge outcomes. □

So the "$(L, R)$-compatible subcategory" of a context-map formalism is not hypothetical. It contains every context map that is injective on predictive atoms: relabellings of latent classes, coarsenings of the observation space that keep the atoms distinct, and their composites.

### 6.5 Typed squares and quintets

**Definition 6.9** (Typed square data)**.** A square with boundary ($h\colon a \to b$, $v\colon a \to c$, $w\colon b \to d$, $k\colon c \to d$) may carry a comparison $\alpha$ only between the composites $k \circ v$ and $w \circ h$, which share source and target. The boundary, the comparison, and the composition laws are all part of the data.

**Proposition 6.10** (Quintets of a locally preordered 2-category)**.** *Let $\mathcal{K}$ be a 2-category whose hom-categories are preorders. The quintet double category $Q(\mathcal{K})$ has the objects and 1-cells of $\mathcal{K}$ in both directions, and exactly one square with boundary $(h, v, w, k)$ iff $k \circ v \leq w \circ h$. Then $Q(\mathcal{K})$ is a strict double category, and interchange, associativity and the unit laws hold automatically.*

*Proof.* This is Ehresmann's quintet construction [10, 12]. Pasting is transitivity together with monotone whiskering, and uniqueness of 2-cells makes every equation trivial. □

In a locally discrete 2-category such as **Meas**, where the only 2-cells are identities [1, §§13.1, 13.5], squares are commuting squares. Nontrivial cells require an order.

### 6.6 What the tower supplies, and what it lacks

(a) *Convex order on one level.* $\leq_{\mathrm{cx}}$ is a partial order on $\mathsf{D}(K)$ for convex $K$, and pushforward along affine maps is monotone. $T \mapsto \mathrm{Law}(S_T Z)$ is a monotone map from the Boolean lattice of leaf sets into $(F_\rho, \leq_{\mathrm{cx}})$. The martingale of theorem 5.3 is a chain of vertical arrows.

(b) *Strict interchange for canonical shadows* (theorem 4.9).

(c) *Lax interchange in general*: a cospan of squares into the meet (corollary 3.8).

(d) *The Beck defects.* The (U) and (M) defects of $\lambda^\sigma$ are vertical arrows of opposite orientation, so $\lambda^\sigma$ is neither lax nor oplax in any $Q(\mathcal{K})$ whose order reads conditional expectation monotonically. The (Cm) defect of $\bar{\lambda}$ is incomparable in the joint convex order [1, Prop. 13.10]. It is a cell of one orientation for label pooling at every level [1, Thm. 13.16].

(e) *What is missing.* $\mathsf{D}$ must act monotonically on hom-preorders. The pointwise convex order fails this, because $\delta_a \leq_{\mathrm{cx}} \delta_b$ forces $a = b$. The candidate is the relation (Barr) lifting of the preorder along $\mathsf{D}$. Since $\mathsf{D}$ preserves weak pullbacks, this lifting preserves reflexivity and transitivity and makes $\mathsf{D}$ monotone.

**Open problem 6.11** (A 2-category for the tower)**.** Are $\mathsf{C}_3$, its structure maps, and the label-pooling cells of $\bar{\lambda}$ monotone for the Barr lifting of label pooling, so that $\bar{\lambda}$ becomes a lax mixed distributive law? This is open problem 4 of [1], now with a definite candidate.

# 7 Beck defects as comparison cells

## 7.1 What a factorisation requires

**Proposition 7.1** (Required data)**.** *To factor a Beck defect through a shadow, one needs:*

*(i) an action of the shadow on the relevant tower objects;*

*(ii) a comparison map through that action;*

*(iii) a typed coherence map completing the factorisation;*

*(iv) a proof that the composite comparison equals the original defect.*

The observation that one side looks pooled and the other unpooled motivates a factorisation but does not prove one. For the defects computed so far, all four items are now supplied.

**Proposition 7.2** (Each computed defect is one conditional-expectation step)**.** *On a fixed labelled space built from the data of the axiom:*

*(a) (U) for $\lambda^\sigma$ at level one: the space is the star space, $Z = \delta_x$, and the step is $E_{\mathrm{triv}}$. The sides are $\mathrm{Law}(E_{\mathrm{triv}}Z) = \delta_\rho$ and $\mathrm{Law}(Z) = \mathsf{D}(\eta)\rho$.*

*(b) (M) for $\lambda^\sigma$ at level one: the space is the sample index, and the step is $E_{\sigma(\ell)}$ (proposition 5.4(b)).*

*(c) (Cm) for $\bar{\lambda}$ at every level and in the shadow: the space is the sample index, and the step is the coordinatewise conditional expectation of the context labels onto the trivial $\sigma$-algebra, with the sample coordinate kept [1, Thm. 13.16].*

*In each case items (i)–(iv) of proposition 7.1 hold: the shadow acts on the labelled space, the comparison is the martingale coupling, the coherence map is the identity on the index, and the equality is the cited computation.*

**Remark 7.3** (The two-shadow conjecture)**.** An earlier formulation conjectured that the (Cm) defect factors through a *pair* of shadows. In every computed case one conditional-expectation step suffices. Whether some Beck defect of a tower variant needs two non-commuting shadows, and so carries a genuinely lax interchange (corollary 3.8), is open.

## 7.2 One-shadow and two-shadow options

Option A keeps one shadow component, $\mathsf{C}_A(Y) = Y \times \mathsf{D}B \times \mathsf{D}^2B \times \mathsf{D}^3B \times \mathsf{Sh}(Y)$. Option B keeps two, $\mathsf{C}_B(Y) = \mathsf{C}_A(Y) \times \mathsf{Sh}(\mathsf{D}Y)$.

**Proposition 7.4.** *Let $\mathsf{C}_E Y = Y \times \prod_i E_i Y$ be built from functors $E_i$, and let $\lambda$ be assembled from transports $\lambda_i \colon \mathsf{D}E_i \Rightarrow E_i\mathsf{D}$. Then (U), (Co) and (M) hold for $\lambda$ iff they hold componentwise. (Cm) does not decompose this way, because $\delta(y, c) = ((y, c), \prod_i E_i(j_c)(c_i))$ feeds the* whole *context into each component.*

*Proof.* The structure maps of $\mathsf{D}$ act factorwise, and equality in a product is componentwise. The formula for $\delta$ is [1, §6]. □

So neither option repairs a failed axiom automatically. For (Cm) the averaging-type transport has its defect in every component at once [1, Thm. 13.16].

### 7.3 Higher cells

**Proposition 7.5** (Triviality after equality)**.** *If two parallel morphisms are equal, the identity comparison on them is trivial. In a quintet double category of a locally preordered 2-category, any two parallel 2-cells are equal, so no nontrivial 3-cells arise.*

A nontrivial 3-cell would require hom-categories that are not preorders. It would then be a comparison between two distinct comparisons with the same boundary. No such structure is used or needed here.

## 8 The measure-theoretic (Giry) extension

Let $X$ be Polish and let $\mathcal{P}(X)$ carry the weak topology. Then $\mathcal{P}$ is a monad on Polish spaces, with $\eta_X(x) = \delta_x$ and $\mu_X(\Pi)(A) = \int \rho(A)\, d\Pi(\rho)$, both continuous [11]. On finite discrete $X$ it is $\mathsf{D}$. A category of closed convex subsets of Euclidean spaces is not closed under $\mathcal{P}$, so the continuous tower must use spaces of measures as objects. A measure-valued shadow needs five ingredients, and each is now available:

(i) *a measurable space of measures*: $\mathcal{P}(X)$ with its Borel structure, which is standard Borel when $X$ is;

(ii) *a conditional barycentre*: for $Z\colon \Omega \to \mathcal{P}(X)$ and $G \subseteq \mathcal{F}$, set $\mathbb{E}[Z \mid G](A) = \mathbb{E}[Z(A) \mid G]$; when $X$ is standard Borel this has a version that is a random probability measure;

(iii) *measurability*: the version can be chosen measurable;

(iv) *integrability*: automatic, since values are probability measures;

(v) *structure*: propositions 4.5 and 4.6 hold verbatim on a fixed probability space.

The commutation results extend as follows.

- (i)⇔(ii)⇔(iii) of theorem 3.3 hold for sub-$\sigma$-algebras of any probability space, applied to $Z(A)$ for $A$ in a countable generating $\pi$-system.
- The finite criteria (iv) and (v) are replaced by the spectral condition $E_A E_B E_A = E_{A\wedge B}$ on $L^2(A)$. The meet need not be countably generated.
- Theorem 3.6 holds with $c$ the Friedrichs cosine. If $c = 1$, the lax interchange holds only in the strong limit.
- For compact metric $X$, proposition 5.2(a,b) and theorem 5.3 hold with $\mathcal{P}(\eta)\rho$ as top. Since $\mathcal{P}(X)$ is a Bauer simplex, maximal representing measures are unique [17].

**Open problem 8.1** (Lax interchange without a rate)**.** When $c = 1$, does the cospan of corollary 3.8 admit a coherent choice, natural along measure-preserving maps that do not pull back the $\sigma$-algebras?

## 9 The finite stochastic-to-convex functor

**Definition 9.1. FinStoch** has as objects finite sets and as morphisms stochastic kernels $K\colon X \to Y$, i.e. $K(y \mid x) \geq 0$ with $\sum_y K(y \mid x) = 1$. Composition is $(L \circ K)(z \mid x) = \sum_y L(z \mid y)K(y \mid x)$, and the identities are Kronecker deltas. It is the Kleisli category of $\mathsf{D}$ restricted to finite sets. **Conv** has as objects the simplices $\Delta_X$ and as morphisms the affine maps. Every kernel induces

$$T_K\colon \Delta_X \to \Delta_Y, \qquad T_K(p)(y) = \sum_x p(x)K(y \mid x). \tag{2}$$

**Lemma 9.2** (Well-definedness and affineness)**.** *$T_K(p) \in \Delta_Y$, and $T_K(tp + (1-t)q) = tT_K(p) + (1-t)T_K(q)$.*

*Proof.* Every term is nonnegative, and $\sum_y T_K(p)(y) = \sum_x p(x) \sum_y K(y \mid x) = 1$. Affineness holds because (2) is linear in $p$. □

**Lemma 9.3** (Identities and composition)**.** *$T_{I_X} = \mathrm{id}$ and $T_{L\circ K} = T_L \circ T_K$.*

*Proof.* $T_{I_X}(p)(x') = \sum_x p(x)\delta_{xx'} = p(x')$. Also $T_{L\circ K}(p)(z) = \sum_x p(x) \sum_y L(z \mid y)K(y \mid x) = \sum_y T_K(p)(y)L(z \mid y)$. □

**Theorem 9.4** (Fully faithful functor)**.** *$T\colon$ **FinStoch** $\to$ **Conv** is a fully faithful functor. Each $T_K$ is continuous and $\ell^1$-non-expansive: $\|T_K(p) - T_K(q)\|_1 \leq \|p - q\|_1$.*

*Proof.* Functoriality is lemmas 9.2 and 9.3. An affine map of simplices is determined by the images of the vertices, which may be any points, so the affine maps $\Delta_X \to \Delta_Y$ are exactly the stochastic matrices. This is the comparison functor from the Kleisli to the Eilenberg–Moore category, restricted to free algebras. Non-expansiveness: $\sum_y |\sum_x (p-q)(x)K(y \mid x)| \leq \sum_x |(p-q)(x)|$ (checks E1, E2). □

**Relation to the tower.** $T$ sends $B$ to $\Delta_B$ and a kernel to an affine map, and a Beck diagram of kernels commutes in **FinStoch** iff it commutes in **Conv**, because $T$ is faithful. Two cautions apply.

- The iterated levels $\mathsf{D}^2B, \mathsf{D}^3B$ are infinite-dimensional convex sets, not finite simplices.
- The shadows are not affine (proposition 5.1(c)), so they are not in the image of $T$, and their interchange is not a commutation of affine maps.

## 10 A Gaussian model: fusion and its type-correct domain

Let $B = \{A, B, C\}$ with uniform prior $\pi$. Let $X_L, X_R$ be real channels, conditionally independent given the class $H = k$:

$$X_L \mid H = k \sim N(\mu_{L,k}, 1), \qquad X_R \mid H = k \sim N(\mu_{R,k}, 1),$$

with $\mu_{L,A} = -2$, $\mu_{L,B} = 0$, $\mu_{L,C} = 2$, $\mu_{R,A} = \mu_{R,C} = -1$ and $\mu_{R,B} = 1$. The left channel separates $A$ from $C$; the right channel cannot. The level-one posteriors $Q_L(x)$, $Q_R(y)$ and $Q_{LR}(x,y)$ take values in $\Delta_B$.

**Proposition 10.1** (Fusion)**.** *$Q_{LR}(x,y) = \mathrm{Fuse}(Q_L(x), Q_R(y))$, where $\mathrm{Fuse}(q,r)_k = \dfrac{q_k r_k/\pi_k}{\sum_j q_j r_j/\pi_j}$.*

*Proof.* $Q_{LR} \propto \pi_k \ell_L(x \mid k)\ell_R(y \mid k)$, and $Q_L Q_R/\pi \propto \pi_k \ell_L(x \mid k)\ell_R(y \mid k)$. This holds exactly under conditional independence (check G1, discrete channels, exact). □

**Domain.** Fuse is defined on $U = \{(q, r) : \sum_j q_j r_j / \pi_j > 0\}$. Gaussian likelihoods are strictly positive, so $Q_L(x)$ and $Q_R(y)$ lie in the open simplex, and $(Q_L, Q_R)$ maps all of $\mathbb{R}^2$ into $U$. The fusion square

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{(Q_L,Q_R)} & U \\ & \searrow^{Q_{LR}} & \downarrow \mathrm{Fuse} \\ & & \Delta_B \end{array}$$

is therefore total and commutes everywhere. No restriction of the domain is needed. The pushforward $(Q_L, Q_R)_* P_{XY}$ has support touching the boundary of $\Delta_B \times \Delta_B$ only in closure, and the boundary has measure zero. The level-two laws are $S_L = (Q_L)_* P_X$, $S_R = (Q_R)_* P_Y$ and $S_{LR} = (Q_{LR})_* P_{XY}$, and the pushforward of the square along $\mathsf{D}$ commutes.

**Shadows in the Gaussian model.** By proposition 3.9, the shadows that commute are $E_{\sigma(X_L,H)}$ and $E_{\sigma(X_R,H)}$, with product $E_{\sigma(H)}$. The shadows $E_{\sigma(X_L)}$ and $E_{\sigma(X_R)}$ do not commute: their meet is trivial, because the joint density is positive, and $X_L$, $X_R$ are dependent through $H$. A relabelling of the latent classes is a measure-preserving bijection that pulls back all three $\sigma$-algebras, so both commuting shadows are natural along it (proposition 4.6).

**What the model shows.** It gives a finite-dimensional, measurable pair of commuting shadows, a total fusion square, and a non-identity natural symmetry. It does not construct the Beck comparison maps. Nor does it show interchange for shadows that omit $H$; that interchange is false.

# 11 Positive tropicalisation and the Berkovich factorisation

## 11.1 Why a real non-Archimedean field

A non-Archimedean absolute value cannot restrict to the usual one on $\mathbb{R}$, so real probability vectors have no informative Berkovich points. For geometry to appear, probabilities must carry *scale*: families $p(t)$ whose coordinates behave like $c\,t^a$ as $t \to 0$.

**Definition 11.1.** $K = \mathbb{R}((t^{\mathbb{R}}))$ is the field of real Hahn series $\sum_{a \in S} c_a t^a$ with $S$ well ordered. It is real closed and complete for $\mathrm{val}(f) = \min S$, with value group $\mathbb{R}$. Write $|f| = e^{-\mathrm{val}\, f}$. $f$ is *positive* if its leading coefficient is positive. The real Puiseux series form a subfield, on which all exact checks are carried out.

The one fact behind everything that follows is this:

$$f, g > 0 \implies \mathrm{val}(f + g) = \min(\mathrm{val}\, f, \mathrm{val}\, g), \qquad \mathrm{val}(fg) = \mathrm{val}\, f + \mathrm{val}\, g. \tag{3}$$

So $\mathrm{val}\colon (K_{>0}, +, \times) \to (\mathbb{R}, \min, +)$ is a semiring homomorphism. This is Maslov dequantisation, the Laplace principle at finite level. A positive $K$-point is a family of distributions together with its large-deviation rates, and val keeps the rates.

## 11.2 The tropical tower

**Definition 11.2** (Two monads)**.** $\mathsf{D}_K X$ is the set of finitely supported $p\colon X \to K_{\geq 0}$ with $\sum p = 1$. $\mathsf{D}_{\mathbb{T}} X$ is the set of finitely supported $v\colon X \to \mathbb{R} \cup \{\infty\}$ with $\min v = 0$, with unit $\eta_{\mathbb{T}}(x) = (0$ at $x$, $\infty$ elsewhere) and multiplication $\mu_{\mathbb{T}}(V)(x) = \min_u (V(u) + u(x))$.

**Theorem 11.3** (Valuation is a monad morphism)**.** $\mathsf{D}_{\mathbb{T}}$ *is a monad, and* $\mathrm{val}\colon \mathsf{D}_K \Rightarrow \mathsf{D}_{\mathbb{T}}$ *is a morphism of monads:*

$$\mathrm{val} \circ \eta = \eta_{\mathbb{T}}, \qquad \mathrm{val}_X \circ \mu_X = \mu_{\mathbb{T},X} \circ \mathrm{val}_{\mathsf{D}_{\mathbb{T}}X} \circ \mathsf{D}_K(\mathrm{val}_X).$$

*Proof.* $\mathsf{D}_{\mathbb{T}}$ is the normalised distribution monad of the semiring $(\mathbb{R} \cup \infty, \min, +)$, and $\min_x \min_u(V(u) + u(x)) = \min V = 0$. The map val lands in $\mathsf{D}_{\mathbb{T}}$ because $\sum p = 1$ forces $\min \mathrm{val}\, p = 0$ by (3). The multiplication square is (3) applied to $\sum_\rho \Pi(\rho)\rho(x)$; atoms merged by $\mathsf{D}_K(\mathrm{val})$ have summed weights, whose valuation is the minimum (checks T1, T2). □

**Corollary 11.4** (The tropical tower)**.** val *maps coherent* $K$*-contexts to coherent tropical contexts,* $e_\perp$ *and* $e_\top$ *to their tropical counterparts, and pooling on a cell to the tropical cell barycentre* $\min_j(\mathrm{val}\, p_j + \mathrm{val}\, Z_j) - \min_j \mathrm{val}\, p_j$.

### 11.3 Fibres

**Definition 11.5.** For positive $\rho \in \mathsf{D}_K X$, the *k-fibre* $F^{(k)}_\rho$ is the set of positive decompositions $\rho = \sum_{i \le k} w_i r_i$. For $v \in \mathsf{D}_{\mathbb{T}} X$, the *tropical k-fibre* $F^{(k)}_{\mathbb{T},v}$ is the set of $(w, u_1, \ldots, u_k)$ with $w \in \mathsf{D}_{\mathbb{T}}[k]$ finite, $u_i \in \mathsf{D}_{\mathbb{T}} X$, and $\min_i(w_i + u_i(x)) = v(x)$ for all $x$.

**Theorem 11.6** (Fibre lifting)**.** $\mathrm{val}(F^{(k)}_\rho) = F^{(k)}_{\mathbb{T},\mathrm{val}\,\rho}$.

*Proof.* $\subseteq$ is theorem 11.3. For $\supseteq$, let $\alpha(x) = \{i : w_i + u_i(x) = v(x)\}$ be the attainers at $x$. For $i \notin \alpha(x)$, choose positive $q_i(x)$ with $\mathrm{val}\, q_i(x) = w_i + u_i(x) > v(x)$. The remainder $m(x) = \rho(x) - \sum_{i \notin \alpha(x)} q_i(x)$ has the leading term of $\rho(x)$; split it equally among the attainers. Then $\sum_i q_i(x) = \rho(x)$ exactly. Put $w_i^K = \sum_x q_i(x)$ and $r_i = q_i / w_i^K$. Then $\mathrm{val}\, w_i^K = w_i + \min u_i = w_i$ and $\mathrm{val}\, r_i = u_i$. Check T3 runs this construction on 300 random points, with 83 tied coordinates. □

**Proposition 11.7.** $F^{(k)}_{\mathbb{T},v}$ *is a finite union of polyhedra, indexed by assignment types* $\alpha\colon X \to 2^{[k]} \setminus \{\emptyset\}$*. It contains the images of the two ends of the classical fibre:* $e_\perp \mapsto (0; v)$ *and* $e_\top \mapsto (w_x = v(x); u_x = \eta_{\mathbb{T}}(x))$.

### 11.4 Tropical secants

**Definition 11.8.** A model $V = \varphi(K^d_{>0}) \subseteq \mathsf{D}_K X$ is *positively parametrised* if the coordinates of $\varphi$ are ratios of polynomials with nonnegative integer coefficients. Its positive tropicalisation is $\mathrm{trop}^+ V = \varphi_{\mathbb{T}}(\mathbb{R}^d)$. The $k$-mixture is $\mathrm{Mix}_k(V) = \{\sum_{i \le k} w_i r_i\}$, the positive part of the $k$-th secant variety, and its tropical analogue is $\mathrm{Mix}^{\mathbb{T}}_k(L) = \{\min_i(w_i + u_i)\}$.

**Theorem 11.9** (Mixtures tropicalise to tropical secants)**.** $\mathrm{val}(V) = \mathrm{trop}^+ V$ *and* $\mathrm{val}(\mathrm{Mix}_k V) = \mathrm{Mix}^{\mathbb{T}}_k(\mathrm{trop}^+ V)$*, exactly. For a toric model* $V_A$*,* $\mathrm{trop}^+ V_A$ *is the row space of* $A$*, normalised, and* $\mathrm{Mix}^{\mathbb{T}}_k$ *is Develin's tropical secant of a linear space [7].*

*Proof.* $\mathrm{val}\, \varphi(\theta) = \varphi_{\mathbb{T}}(\mathrm{val}\, \theta)$ by (3), and val maps $K^d_{>0}$ onto $\mathbb{R}^d$. For mixtures, apply theorem 11.3 and choose lifts with prescribed valuations. □

Over an algebraically closed field, cancellation can make $\mathrm{trop}(\mathrm{Sec}_k)$ larger than the tropical secant [7, 8]. On the positive part there is no discrepancy. For matrices, the positive tropicalisation of the nonnegative-rank-$\le k$ locus is the Barvinok rank-$\le k$ locus. For real semialgebraic sets the image of val is Alessandrini's logarithmic limit set [4].

**Example 11.10** (Dimensions)**.** Generic Jacobian ranks (floating point, tolerance $10^{-6}$, 40 points; checks T6, T7):

| model | $k$ | $\dim \mathrm{Mix}_k^{\mathbb{T}}$ | $\dim \mathrm{Mix}_k$ (affine cone) |
|---|---|---|---|
| $3 \times 3$ independence | 1, 2, 3 | 5, 8, 9 | 5, 8, 9 |
| $2 \times 2 \times 2$ independence | 1, 2 | 4, 8 | 4, 8 |

In each case the tropical dimension certifies the classical one, in the spirit of [9].

## 11.5 The Berkovich square

$(\mathbb{A}^m_K)^{\mathrm{an}}$ is the space of multiplicative seminorms on $K[z]$ extending $|\cdot|$ [6]. A $K$-point $y$ gives the rigid point $|f|_y = |f(y)|$. On the torus there are two maps:

- the skeleton section $s(v)$, the Gauss point $|\sum_m a_m z^m|_{s(v)} = \max_m |a_m| e^{-\langle m, v\rangle}$;
- the retraction $\tau(\xi) = (-\log |z_j|_\xi)_j$.

They satisfy $\tau s = \mathrm{id}$, and $\tau(y) = \operatorname{val} y$ on rigid points. Payne's theorem [16] identifies the analytification with the inverse limit of tropicalisations, and Jell, Scheiderer and Yu prove the real analogue for semialgebraic sets [13].

**Theorem 11.11** (Berkovich square)**.** *Let $\Phi\colon \mathbb{A}^m \to \mathbb{A}^n$ be polynomial with integer coefficients, for instance the $k$-mixture map, and assume the residue characteristic is $0$. Then:*

*(a) $\tau \circ \Phi^{\mathrm{an}} \circ s = \Phi_{\mathbb{T}}$ on the skeleton;*

*(b) $\tau \circ \Phi^{\mathrm{an}} = \Phi_{\mathbb{T}} \circ \tau$ on positive rigid points, if $\Phi$ has nonnegative coefficients;*

*(c) $\tau \circ \Phi^{\mathrm{an}} \geq \Phi_{\mathbb{T}} \circ \tau$ on all rigid points, strictly exactly where leading terms cancel.*

$$\begin{array}{ccc} F_\rho^{(k)} \subseteq (\mathit{params})_K^{>0} & \xrightarrow{\ \Phi\ } & \mathrm{Mix}_k(V) \subseteq \mathsf{D}_K X \\ \Big\downarrow{\scriptstyle \mathit{rigid}} & & \Big\downarrow{\scriptstyle \mathit{rigid}} \\ (\mathbb{A}^m)^{\mathrm{an}} & \xrightarrow{\ \Phi^{\mathrm{an}}\ } & (\mathbb{A}^n)^{\mathrm{an}} \\ {\scriptstyle s}\big(\ {\scriptstyle \tau}\big\downarrow & & \Big\downarrow{\scriptstyle \tau} \\ F_{\mathbb{T},v}^{(k)} \subseteq \mathbb{R}^m & \xrightarrow[\ \Phi_{\mathbb{T}}\ ]{} & \mathrm{Mix}_k^{\mathbb{T}} \subseteq \mathsf{D}_{\mathbb{T}} X \end{array}$$

*Proof.* (a) $-\log |\Phi_j|_{s(v)} = \min_m \langle m, v\rangle$, because nonzero integers have absolute value 1 (check T4). (b) is (3). (c) is the ultrametric inequality. □

## 11.6 What the analogies do not give

A Berkovich point is a multiplicative seminorm, not a probability measure. A rational domain is an algebraic localisation, not conditioning. The Kleisli category is not a site without specified covers. A distributive law is not a structure sheaf. Shadows are conditional expectations, not Grothendieck topologies. Two constructions that tensor $C(\Delta_X)$ with a non-Archimedean field fail for definite reasons:

- $\mathbb{R}$ has no isometric embedding into such a field;

- $C(\Delta_X)$ is not affinoid;
- over $\mathbb{R}$, Gel'fand duality returns $\Delta_X$ and the functor $T$ of theorem 9.4, with nothing new.

What *does* hold is theorem 11.11. Positivity, through (3), is the hypothesis that turns the analogy into a theorem.

# 12 Shadows at zero temperature and secant geometry

## 12.1 The posterior tropicalises to the hard assignment

**Proposition 12.1.** *For positive $\rho = \sum_i w_i r_i$, the posterior $P(i \mid x) = w_i r_i(x)/\rho(x)$ has* $\operatorname{val} P(i \mid x) = w_i^{\mathbb{T}} + u_i(x) - v(x) \geq 0$, *with equality iff $i$ attains the minimum. For convergent series, its $t \to 0$ limit is* $\operatorname{lc}(w_i r_i(x))/\operatorname{lc}(\rho(x))$ *on the attainers and* $0$ *elsewhere, where* lc *is the leading coefficient (check T5).*

So conditioning on the observation tropicalises to argmin (MAP) assignment. Nearest-atom pooling is a *zero-temperature shadow*, and its Voronoi walls are tie loci. The same mechanism gives $k$-means as the small-variance limit of Gaussian mixtures [15].

**Proposition 12.2** (Walls)**.** *(a) $\mu_{\mathbb{T}}$ is affine on each assignment cell. Crossing a tie hyperplane $w_i + u_i(x) = w_j + u_j(x)$ changes the assignment and the affine piece (check T8).*

*(b) If $r_i = r_j$, the classical fibre contains the weight-split segment $\{w_i + w_j = c\}$. Its valuation image is the bent line $\{\min(w_i^{\mathbb{T}}, w_j^{\mathbb{T}}) = \operatorname{val} c\}$, made of two rays.*

## 12.2 Statistical models as varieties

A parametric model is algebraic if $\theta \mapsto \mathbb{E}_\theta[s(X)]$ is polynomial or rational for a vector of sufficient statistics $s$. Its image is the *model variety*. For $N(\mu, \Sigma)$ on $\mathbb{R}^d$ with moments up to order $r$, the image is the Gaussian moment variety $G_{d,r}$, of dimension $d + d(d+1)/2$ when the parametrisation is generically injective. A mixture of $k$ Gaussians has moment vector $\sum_i p_i m_r(\mu_i, \Sigma_i)$, so its image is the positive part of $\operatorname{Sec}_k(G_{d,r})$ [3]. Its parameter count is

$$\dim \Theta_{d,k} = k\big(d + \tfrac{d(d+1)}{2}\big) + k - 1,$$

and its expected dimension is $\min\{k \dim G_{d,r} + k - 1,\ \text{ambient dimension}\}$. The model is *generically identifiable* when the mixture map is generically finite-to-one, which requires $\dim \operatorname{Mix}_k = \dim \Theta_{d,k}$. It is *defective* when $\operatorname{Sec}_k$ has smaller than expected dimension. Defectivity is a global property of $\operatorname{Sec}_k$, not a locus.

**Proposition 12.3** (Moment order and the tower)**.** *(a) The order-$r$ moment map of a two-component univariate Gaussian mixture has generic Jacobian rank* $\min(r, 5)$. *At order two the single-Gaussian image $\{(m, v + m^2)\}$ is already open, so second moments cannot see mixtures, and local identifiability needs $r \geq 5$ (check D1).*

*(b) Affine statistics are constant on barycentre fibres. Regrouping a mixture of mixtures leaves the flattened law and all its moments unchanged (check D2). Hence the moment image of every tower level equals that of its flattening. If every node has $m$ children, level $n$ lies in* $\operatorname{Mix}_{m^n}$, *and the hierarchy itself is invisible to secant varieties. It lives in the convex-order fibre (proposition 5.2).*

*Proof.* (a) is direct computation at a rational point. (b) Moments are integrals, which are affine in the measure, and $\mu$ preserves integrals of affine functions. □

### 12.3 Walls of a component collision

For $\nu_\varepsilon = p\,N(\mu,\Sigma) + (1-p)\,N(\mu+\varepsilon v,\Sigma)$ with $0<p<1$, $\Sigma \succ 0$ and $v \neq 0$, the two-component fibre over $\nu_\varepsilon$ is finite for $\varepsilon \neq 0$, up to label swap. At $\varepsilon = 0$ it becomes the segment $\{(p_1,p_2;\mu,\Sigma) : p_1+p_2=1\}$. The collision locus is $\mathrm{Mix}_{k-1} \subset \mathrm{Mix}_k$, where the $k$-component parametrisation has positive-dimensional fibres. Tropically, by proposition 12.2(b), the collision is the locus $u_i = u_j$, where the tropical fibre gains a bent line.

**Conjecture 12.4** (Wall-crossing, proposed)**.** *For a path $\nu_t$ crossing the collision locus transversally at $\nu_0$, the change in the fibre is a finite sum $\sum_\sigma c_\sigma F_\sigma(\nu_0)$ over the collision strata $\sigma$ through $\nu_0$, with combinatorial coefficients $c_\sigma$ determined by the local structure of the collision. In the single-collision example, $c_\sigma = 1$ and $F_\sigma$ is the weight-split segment.*

The formal template is the wall-crossing of enumerative geometry, where the difference across a wall is a sum over strata. This is an analogy, not an established correspondence. The tropical picture of proposition 12.2 offers a concrete first test: the change in the tropical fibre across a tie or collision locus is a polyhedral computation.

**Open problem 12.5** (Tropical identifiability)**.** For $\rho \in \mathrm{Mix}_k(V)$, is $\mathrm{val}(F_\rho^{(k)} \cap D_k(V)) = F^{(k)}_{\mathbb{T},\mathrm{val}\,\rho} \cap D_k^{\mathbb{T}}(\mathrm{trop}^+ V)$? For $V = \mathsf{D}_K X$ this is theorem 11.6. Lifting failures inside $V$ would be tropical certificates of non-identifiability.

**Open problem 12.6** (Order on tropical fibres)**.** Is the transitive closure of tropical pooling a partial order on $F_{\mathbb{T},v}$, with the two points of proposition 11.7 as bottom and top, and does it orient the tropical Beck defects of $\lambda^\sigma$ as proposition 5.4 does classically?

**Open problem 12.7** (Čech classes)**.** No construction is proposed here that represents Beck defects as Čech cocycles on a skeleton. One would need a site, a coefficient sheaf, and a map from defects to cochains whose vanishing is equivalent to factorisation. The tropical walls of proposition 12.2 are the natural place to look for covers.

## 13 Computational checks

### 13.1 The two scripts

`check_shadows.py` has 30 checks: exact rational arithmetic, except A1(v) and A4, which use double precision with tolerance $10^{-9}$. Tower objects are imported from `check_tower.py`. `check_tropical.py` has 8 checks: exact Puiseux arithmetic, except the floating-point ranks in T6 and T7. All pass; the hashes are in `FROZEN.txt`.

| | |
|---|---|
| A1–A7b | theorem 3.3 (449 commuting, 151 non-commuting); proposition 3.4; example 3.5; theorem 3.6; corollary 3.8; lemma 3.2; proposition 3.9 |
| B1–B7 | star space; theorem 4.9; proposition 4.10; theorem 5.3; proposition 4.11; propositions 5.1 and 5.2; proposition 5.4 |
| C1–C7 | audit of earlier scripts; proposition 4.2; example 4.3; proposition 5.5(b) |
| D1–D2, E1–E2 | proposition 12.3; theorem 9.4 |
| G1–G3 | proposition 10.1; proposition 4.6; proposition 6.8 |
| T1–T8 | theorem 11.3; theorem 11.6; theorem 11.11(a); proposition 12.1; example 11.10; proposition 12.2(a) |

### 13.2 What checks establish

Finite exact checks verify identities on the enumerated inputs and can falsify overbroad claims. They do not prove identities for all measures, establish measurability, or supply categorical typing. Every general statement in this paper has a proof; the checks guard the proofs against slips. Hashes certify the identity of artefacts, not the truth of theorems.

### 13.3 Audit of earlier scripts

Five scripts accompanied an earlier version. All five share one nearest-atom `pooling(nu, atoms)`, with "type-aware" total variation and ties broken by key order. They pass every reported check for the following reasons:

| Script | Why it passes |
|---|---|
| `check_giry.py` | $\Pi$ always has two atoms, and $\Sigma$'s descriptors $i$ mod 2 biject with them. So $\sigma(\theta) = \sigma(\tilde{r})$, and $L$ and $R$ have identical cells (C1). |
| `check_3cat-Coherence.py` | As above. In addition, level-5 atoms of type $\mathsf{D}^2$ against inputs of type $\mathsf{D}^1$ are all at TV distance 1, so the pooling is total, $e_\perp$, and the "3-cell" is trivial (C7). |
| `check_non-coherent.py` | The duplicated predictive $(0, \rho_1), (1, \rho_1)$ ties, and the first key wins, so $R$'s cells equal $L$'s (C6). |
| `double_category.py` | "Interchange" is $R(L\nu) = L(R\nu)$ with $L = R$. "(U) defect nonzero" is $e_\perp \neq e_\top$. |
| `check_span.py` | The cells coincide (C1). `dist_of` strips $\theta$, so "not inverse" holds by a change of key type. "(Cm) equals sum" compares a pair with itself (C5). Mixed inputs combine two key types under one metric. |

By proposition 4.2, any composite of nearest-atom poolings is one pooling onto a coarsening of the first partition. By proposition 12.1, such pooling is a zero-temperature shadow. The scripts' counts therefore carry no information about theorem 3.3. A meaningful test uses the star space with labels that are not functions of the predictive, and compares three pairs: canonical shadows, label-only shadows, and naive composites. Checks B2, B3 and C2 do exactly this.

## 14 Consequence for the tower paper

The tower paper [1] never uses $E_B E_A = E_{A \vee B}$. Its poolings live on a fixed sample index [1, Def. 13.13], and its order in [1, Thm. 13.16] is a transitive closure. An unnumbered paragraph added after that theorem explains why the closure is needed. A two-step label pooling $E_{G_2} E_{G_1}$ is one conditional expectation iff the two commute, which by theorem 3.3 means conditional independence given $G_1 \wedge G_2$; the product is then $E_{G_1 \wedge G_2}$, never the join.

## 15 Summary and open problems

- Two conditional expectations commute iff they are conditionally independent given their meet. Otherwise they interchange laxly, through the meet, at the exact rate $c^{2n-1}$. The join identity is false except for equal $\sigma$-algebras.

- The canonical shadows of a context live on its star space. They form a commutative band, are natural along context maps that are injective on predictive atoms, and interchange strictly.
- The tower acts on barycentre fibres as a martingale. $e_\perp$ and $e_\top$ are the bottom and top of each fibre. Each computed Beck defect is one conditional-expectation step.
- Over a real non-Archimedean field the whole picture tropicalises exactly: the valuation is a monad morphism, fibres lift, mixtures become tropical secants, and the valuation factors through Berkovich spaces. Hard assignment is the zero-temperature shadow.

The open problems are open problems 6.11, 8.1 and 12.5 to 12.7, and conjecture 12.4.

# A Changes from the previous versions

This paper supersedes three manuscripts of October 2026:

(i) *Conditional Expectations, Beck Defects, and a Proposed Double-Category Structure*, referred to below as G;

(ii) *Conditional Expectations, Beck Defects, and the Shadow Algebra of the Posterior Tower*, referred to below as S;

(iii) *From Barycentre Fibres to Secant Geometry.*

S and (iii) are absorbed in full. The table lists every statement of G that changed.

| G | Status | Here |
|---|---|---|
| Def 3.2, Props 3.3, 3.5 (label shadows) | kept | proposition 4.1; composites in proposition 4.2 |
| Thm 4.2 (nested) | kept | special case of theorem 3.3 |
| Thm 4.4 (CI $\Rightarrow$ commute) | strengthened | five-way equivalence, theorem 3.3 |
| Rem 4.5 (join), Ex 4.6 | kept, sharpened | proposition 3.4; example 3.5 |
| Rem 4.7 ("hypotheses necessary") | made precise | theorem 3.3; general case theorem 3.6 and corollary 3.8 |
| Prop 4.8, Rem 4.9, Prop 8.1 (inheritance) | discharged | proposition 4.5 |
| Defs 5.1, 5.3, Rem 5.4, Defs 5.5, 5.6, Props 5.7, 5.8, Rem 5.9 | kept | section 6; context maps defined concretely in definition 6.1 |
| Def 5.10, Prop 5.11 (compatible subcategory) | proved | propositions 4.6 and 6.8 |
| Diagram after Def 5.13 | withdrawn | horizontal composition of contexts is not a functor $\mathbf{Ctx} \times \mathbf{Ctx} \to \mathbf{Ctx}$ |
| Thm 5.14 (interchange blueprint) | replaced | proposition 6.10 |
| Prop 6.1 (required data) | kept | proposition 7.1, now supplied by proposition 7.2 |

| G | Status | Here |
|---|---|---|
| Conj 6.2 (two-shadow factorisation) | superseded | one step suffices in every computed case |
| Thms 7.1, 7.2 (Options A, B) | partly invalid | proposition 7.4: (Cm) does not decompose |
| §8 (Giry) | completed | section 8 |
| Prop 9.1, Rem 9.2 (3-cells) | kept | triviality in quintets |
| §10 (regimes) | kept | worked regimes in section 3 |
| Def 11.1, Thms 11.2, 11.5 (double category of the tower) | withdrawn | 2-cells mistyped; proposition 6.10 |
| Thm 11.4, Prop 11.8 | kept | identities and latent relabellings are natural (section 10) |
| Prop 11.7 (Gaussian interchange) | false | proposition 3.9 |
| §11.6.1 (restricted domain $V$) | corrected | the square is total (section 10) |
| §12 (scope of checks) | kept | section 13 |
| §13 (Berkovich programme) | replaced | section 11 |
| §14 (stochastic-to-convex bridge) | kept, strengthened | theorem 9.4: fully faithful; shadows not affine |
| §15 (Wasserstein–Berkovich functor) | withdrawn | ill-defined; theorem 11.11 instead |
| §16 (secants, wall-crossing) | corrected | section 12: order $\geq 3$; hierarchy invisible to moments; conjecture 12.4 |

## References


[1] V. K. Awasthi, The three-level posterior tower: hierarchical Bayesian inference and mixed distributive laws. Technical report, October 2026 (version 10).

[2] V. K. Awasthi, Pooling idempotents and the Beck defects of the absolute tower. Note, October 2026.

[3] C. Améndola, J.-C. Faugère and B. Sturmfels, Moment varieties of Gaussian mixtures. *J. Algebr. Stat.* **7** (2016), 14–28.

[4] D. Alessandrini, Logarithmic limit sets of real semi-algebraic sets. *Adv. Geom.* **13** (2013), 155–190.

[5] H. H. Bauschke, F. Deutsch and H. Hundal, Characterizing arbitrarily slow convergence in the method of alternating projections. *Int. Trans. Oper. Res.* **16** (2009), 413–425.

[6] V. G. Berkovich, *Spectral Theory and Analytic Geometry over Non-Archimedean Fields*. AMS, 1990.

[7] M. Develin, Tropical secant varieties of linear spaces. *Discrete Comput. Geom.* **35** (2006), 117–129.

[8] M. Develin, F. Santos and B. Sturmfels, On the rank of a tropical matrix. In *Combinatorial and Computational Geometry*, MSRI Publ. 52, 2005, 213–242.

[9] J. Draisma, A tropical approach to secant dimensions. *J. Pure Appl. Algebra* **212** (2008), 349–363.

[10] C. Ehresmann, Catégories structurées. *Ann. Sci. Éc. Norm. Sup.* **80** (1963), 349–426.

[11] M. Giry, A categorical approach to probability theory. LNM 915, Springer, 1982, 68–85.

[12] M. Grandis, *Higher Dimensional Categories: From Double to Multiple Categories*. World Scientific, 2019.

[13] P. Jell, C. Scheiderer and J. Yu, Real tropicalization and analytification of semialgebraic sets. *Int. Math. Res. Not.* (2022), 928–958.

[14] S. Kayalar and H. L. Weinert, Error bounds for the method of alternating projections. *Math. Control Signals Systems* **1** (1988), 43–59.

[15] B. Kulis and M. I. Jordan, Revisiting $k$-means: new algorithms via Bayesian nonparametrics. *ICML* 2012.

[16] S. Payne, Analytification is the limit of all tropicalizations. *Math. Res. Lett.* **16** (2009), 543–556.

[17] R. R. Phelps, *Lectures on Choquet's Theorem*, 2nd ed. LNM 1757, Springer, 2001.

[18] V. Strassen, The existence of probability measures with given marginals. *Ann. Math. Statist.* **36** (1965), 423–439.

[19] J. von Neumann, *Functional Operators, Vol. II*. Princeton University Press, 1950.